

Name: _____ Date: _____

DECIMALS

Decimal Place Value Chart:

Hundreds | Tens | Ones . Tenths | Hundredths | Thousandths

1 3 7 . 4 0 5

Read as: "one hundred thirty-seven AND four hundred five thousandths" (137.405)

Key: tenths = $\frac{1}{10}$, hundredths = $\frac{1}{100}$, thousandths = $\frac{1}{1000}$

Part A – Name the Place & Value

Number	Underlined digit	Place Name	Value
4 <u>7</u> .305	7		
12.4 <u>8</u> 6	8		
0. <u>0</u> 53	0		
304.0 <u>9</u> 1	9		
5.7 <u>3</u> 8	3		

Part B – Write in Words and Standard Form

1. Write 8.047 in words:

2. Write 0.305 in words:

3. "Fourteen and sixty-two thousandths" =

4. "Three hundred and eight tenths" =

Part C – Decimals as Fractions

5. $0.7 = \frac{\square}{\square}$

6. $0.43 = \frac{\square}{\square}$

7. $0.125 = \frac{\square}{\square}$

8. $\frac{9}{100} = \square$

Part D – Compare & Order Decimals

Line up the decimal points! Compare digit by digit from left to right.

0.06 0.60 0.600

10.245 2.450 2.405

Name: _____ Date: _____

ROUNDING DECIMALS

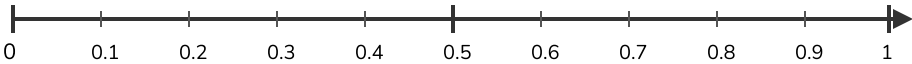
Rounding decimals — same rule as whole numbers:
Look at the digit ONE place to the RIGHT of where you're rounding.
0–4 → round down, 5–9 → round up.
Round 4.376 to the nearest tenth: look at hundredths digit ($7 \geq 5$) → round up → **4.4**
Round 4.376 to the nearest hundredth: look at thousandths digit ($6 \geq 5$) → round up → **4.38**

Part A – Round the Decimals

Number	Nearest whole	Nearest tenth	Nearest hundredth
6.382			
0.547			
14.905			
2.999			
0.0047			

Part B – Number Line: Locate the Decimal

1. Mark and label 0.3, 0.7, and 0.15 on the number line below.



2. What decimal is exactly halfway between 0.4 and 0.5?

3. What decimal is exactly halfway between 3.6 and 3.7?

Part C – Real-World Rounding

4. A bag of chips weighs 8.637 oz. Round to the nearest tenth:

5. Gas in Peoria costs \$3.879 per gallon. A news report rounds to the nearest cent (hundredth). What does the report say?

6. A runner's time is 9.843 seconds. Rounded to the nearest tenth, was it faster or slower than 9.9 seconds?

Summer Math Adventure

W4
D3

Week 4 · Day 3 | Adding & Subtracting Decimals

Name: _____

Date: _____

DECIMAL OPERATIONS

The golden rule: LINE UP THE DECIMAL POINTS!

Then add zeros to make the same number of decimal places. Carry and borrow just like whole numbers.

Example: $12.4 + 3.067$

12.400

+ 3.067

15.467 ← decimal point stays in same column!

Part A – Set It Up & Solve

Rewrite vertically with decimal points aligned, add zeros where needed.

1. $4.3 + 2.85 =$

2. $13.7 + 0.048 =$

3. $8.4 - 3.27 =$

4. $10 - 4.375 =$

5. $6.003 + 14.97 + 0.8 =$

6. $25.06 - 18.4 =$

Part B – Error Hunt

7. A student says: $3.4 + 0.06 = 3.46$. Is this right? Circle: YES / NO

If wrong, what did they do wrong?

Correct answer:

8. A student says: $5 - 1.2 = 4.2$. Is this right? Circle: YES / NO

Correct answer:

Summer Math Adventure

W4
D4

Week 4 · Day 4 | Multiplying Decimals

Name: _____

Date: _____

MULTIPLYING DECIMALS

How to multiply decimals:

Step 1: Ignore the decimal points and multiply as whole numbers.

Step 2: Count total decimal places in BOTH numbers.

Step 3: Place the decimal point that many places from the RIGHT in your answer.

Example: $3.4 \times 0.7 \rightarrow 34 \times 7 = 238 \rightarrow 2$ decimal places total $\rightarrow$ **2.38**

Sense check: $3.4 \times 0.7 \approx 3 \times 0.7 = 2.1$. Answer 2.38 is close. ✓

Part A – Multiply (Count the decimal places!)

1. $0.4 \times 0.3 =$

Whole: $4 \times 3 =$ ____

Decimal places: ____

Answer:

2. $1.2 \times 0.5 =$

Whole: $12 \times 5 =$ ____

Decimal places: ____

Answer:

3. $2.4 \times 1.3 =$

Whole: $24 \times 13 =$ ____

Decimal places: ____

Answer:

4. $0.08 \times 0.7 =$

Answer:

5. $3.14 \times 2.5 =$

Answer:

6. $0.25 \times 0.04 =$

Answer:

Part B – Decimal $\times$ Whole Number

7. $3.6 \times 7 =$

8. $12.4 \times 5 =$

9. $0.045 \times 8 =$

Part C – Powers of 10 Shortcut

Multiplying by 10, 100, 1000? Move the decimal RIGHT. By 0.1, 0.01? Move it LEFT.

10. $4.7 \times 10 =$

11. $4.7 \times 100 =$

12. $4.7 \times 0.1 =$

13. $4.7 \times 1,000 =$

14. $4.7 \times 0.01 =$

15. $0.006 \times 10 =$

16. A car gets 32.4 miles per gallon. The gas tank holds 12.5 gallons. How far can the car travel on a

Summer Math Adventure

W4
D5

Week 4 · Day 5 | Dividing Decimals & Cumulative Review

Name: _____

Date: _____

DIVIDING DECIMALS

Dividing a decimal by a whole number: Place decimal in quotient directly above the decimal in the dividend.

Divide as usual.

$$6.48 \div 4 = 1.62 \quad (\text{think: } 648 \div 4 = 162, \text{ then place decimal: } 1.62)$$

Dividing by a decimal: Multiply BOTH numbers by 10/100 to make divisor a whole number.

$$3.6 \div 0.4 = 36 \div 4 = 9 \quad (\text{multiply both by } 10)$$

Part A – Decimal ÷ Whole Number

1. $8.4 \div 7 =$

2. $14.4 \div 6 =$

3. $0.96 \div 8 =$

4. $32.5 \div 5 =$

5. $7.35 \div 3 =$

6. $4.56 \div 12 =$

Part B – Decimal ÷ Decimal

Multiply both numbers to eliminate the decimal in the divisor first.

7. $2.4 \div 0.6 =$

→ $__\div__ =$

8. $3.5 \div 0.7 =$

→ $__\div__ =$

9. $0.36 \div 0.04 =$

→ $__\div__ =$

Part C – Word Problems

10. You and 4 friends split a restaurant bill of \$48.75 equally. How much does each person pay?

Each pays: \$

11. A rope is 13.5 meters long. It is cut into pieces of 0.9 meters each. How many pieces are there?

Answer:

Week 4 Quiz – Decimals

W4
QUIZ

Topics: Place Value · Rounding · Add/Subtract · Multiply · Divide

Name: _____

Date: _____

Score: _____ / 20

Section 1 – Place Value & Rounding (5 pts)

1. What is the value of 4 in 12.346?

2. Write 0.305 as a fraction.

3. Order: 0.5, 0.05, 0.51, 0.505 (least to greatest)

4. Round 7.4852 to nearest hundredth:

5. **(1 pt)** Write in standard form: "fifty-two and seven hundredths" =

Section 2 – Add & Subtract (5 pts)

6. $8.4 + 3.076 =$

7. $20 - 7.38 =$

8. **(3 pts)** Mei ran 3.75 km on Monday, 4.2 km on Wednesday, and 2.875 km on Friday. How many total km did she run? Round your answer to the nearest tenth.

Total:

km Rounded:

km

Section 3 – Multiply & Divide (10 pts)

9. $2.4 \times 3.5 =$

10. $0.07 \times 0.9 =$

11. $6.48 \div 4 =$

12. $4.5 \div 0.9 =$

13. **(2 pts)** A bag of trail mix weighs 1.35 kg. There are 6 bags. What is the total weight?
